AMERICAN ACADEMY OF OPHTHALMOLOGY®

Prediction Models for Retinopathy of Prematurity Using Nonimaging Machine Learning Approaches: A Regional Multicenter Study

Yusuke Takeda, MD, MPH,[1] *Yutaka Kaneko, MD, PhD,*[1] *Masahiko Sugimoto, MD, PhD,*[1] *Hidetoshi Yamashita, MD, PhD,*[2] *Ayako Sasaki, MD, PhD,*[3] *Tetsuo Mitsui, MD, PhD*[3]

Purpose: To develop nonimaging machine learning models using clinical data from the first screening to predict the occurrence of retinopathy of prematurity (ROP).

Design: This multicenter regional study was conducted in Yamagata Prefecture, Japan.

Participants: We collected clinical data of neonates born between October 2016 and September 2018 and screened in 4 neonatal care units.

Methods: The 35 variables available at the first screening were used as possible predictors to develop a decision tree, a random forest, a gradient-boosted tree, a neural network, and a Naive Bayes model. Parameter tuning was performed using a 10-fold cross-validation. This process was repeated 200 times using different random seeds for data partitioning.

Main Outcome Measures: The target outcome was the final ROP outcome (i.e., the development of any stage of ROP during hospitalization).

Results: Of the 215 neonates screened, 43 (20.0%) developed ROP. The median gestational age was 31.4 (interquartile range: 28.1–33.4) weeks, and the median birth weight was 1502 (interquartile range: 967–1823) g. The mean 200-iteration area under the receiver operating characteristic curve (AUC-ROC), accuracy, sensitivity, and specificity of the random forest model were 0.93 (95% confidence interval [CI] 0.83–0.99), 90.1% (95% CI 84.1–95.2), 95.7% (95% CI 88.2–100), and 66.0% (95% CI 41.7–91.7), respectively. The mean 200-iteration AUC-ROC, accuracy, sensitivity, and specificity of the Naive Bayes model were 0.94 (95% CI 0.86–0.99), 90.6% (95% CI 84.1–96.8), 94.6% (95% CI 86.3–100), and 73.6% (95% CI 50.0–91.7), respectively.

Conclusions: Nonimaging machine learning methods have shown high performance in predicting the occurrence of ROP. These models can be beneficial when a fundus camera cannot capture images due to eye opacity and for hospitals that lack pediatric fundus cameras.

Financial Disclosure(s): The author(s) have no proprietary or commercial interest in any materials discussed in this article. *Ophthalmology Science 2025;5:100715 © 2025 by the American Academy of Ophthalmology. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).*

Retinopathy of prematurity (ROP) is the primary cause of childhood blindness, with increasing worldwide incidence.[1–3] Advancements in perinatal medicine have made it possible to save neonates who were previously unable to survive, consequently reducing the risk of developing ROP at the population level.[3,4] In practice, fundus examinations are initiated based on the screening criteria, mainly comprising gestational age and birth weight. Repeated fundus examinations are needed for optimal diagnosis and treatment; however, they physically burden neonates.

Although there are many research publications on prediction models for progression to severe ROP or treatment-warranted ROP,[5–9] there are only a few reports on prediction models for its occurrence. Regarding the occurrence of ROP, scoring-based and logistic regression models using clinical status showed moderate predictive performance.[10–13] A deep learning model using fundus images reported good predictive ability[14]; however, highly premature neonates cannot have fundus photographs taken, and some hospitals cannot afford pediatric fundus cameras. Despite the variety of models, there remains a gap in research publications concerning the prediction models for ROP occurrence. To the best of our knowledge, no publication has been published on machine learning models using only tabular data. Such data, which include demographic and clinical information, are more readily available than imaging data obtained using pediatric fundus cameras, especially in low-resource settings.

This study aimed to develop prediction models for the occurrence of ROP using nonimaging machine-learning approaches during the first screening examination. These

© 2025 by the American Academy of Ophthalmology
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/). Published by Elsevier Inc.

https://doi.org/10.1016/j.xops.2025.100715
ISSN 2666-9145/25

prediction models may be beneficial when prediction using fundus camera data is not possible, such as in neonates who cannot be photographed because of eye conditions or in hospitals not equipped with pediatric fundus cameras.

Methods

Study Design and Population

This was a regional, multicenter study conducted in Yamagata Prefecture, Japan. The clinical data of neonates born between October 2016 and September 2018 and screened in all 4 neonatal care units (NICUs) in the area were acquired (Yamagata University Hospital, Yamagata Prefectural Central Hospital, Yamagata Saisei Hospital, and Tsuruoka Municipal Shonai Hospital). Data were collected prospectively to investigate descriptive epidemiology, risk analyses, and prediction models. This study retrospectively analyzed the collected data to develop predictive models using machine learning methods. Written informed consent was obtained from the parents or guardians of patients. Neonates were screened according to the Japanese criteria: birth weight $\leq$1800 g, gestational age $\leq$34 weeks, or neonates at risk despite exceeding these thresholds.[15] Fundus examinations commenced at 31 weeks of postmenstrual age for neonates born before 28 weeks of gestation and at 4 weeks of age for neonates born at or after 28 weeks of gestation.[16] The diagnosis was determined based on the International Classification of ROP Revisited,[17] and treatment was performed according to the report from the Early Treatment for ROP Study.[18] Binocular indirect ophthalmoscopy and 20 or 28 diopter lenses were used for the fundus examination. All fundus examinations and treatments were performed by an ophthalmologist board certified by the Japanese Ophthalmological Society. Data were collected on the final ROP outcome, gestational age, birth weight, and other neonatal factors related to treatment and comorbidities. Maternal and other related factors described in neonatal medical records were also used. This study was approved by the Ethical Review Committee of Yamagata University Faculty of Medicine (No. 2016-244) and conformed to the tenets of the Declaration of Helsinki.

Target Outcome and Predictors

The target outcome was the final ROP outcome, that is, whether the neonates developed ROP during hospitalization. Targeting each stage of ROP results in a lower occurrence level; therefore, any stage of ROP was used to build robust predictive models. We classified neonates with any stage of ROP as "Any ROP" and those without ROP as "No ROP."

The 35 variables available at the first screening were used as possible predictors: gestational age, birth weight, body weight at the first screening, intrauterine growth retardation, female sex, 1-minute appearance, pulse, grimace, activity, respiration (Apgar) score, 5-minute Apgar score, endotracheal intubation, surfactant administration, respiratory distress syndrome, transient tachypnea of the newborn, caffeine administration, chronic lung disease, hydrocephalus, periventricular leukomalacia, intraventricular hemorrhage, sepsis, necrotizing enterocolitis, patent ductus arteriosus, patent ductus arteriosus treated with indomethacin, patent ductus arteriosus treated with surgery, erythropoietin administration, blood transfusion, and human milk feed as neonatal factors; maternal age, placental abruption, premature rupture of the membranes, placenta previa, gestational diabetes mellitus, pregestational diabetes mellitus, pregnancy-induced hypertension, chorioamnionitis, and steroid administration at delivery as maternal factors; and multiple birth and caesarean section as other factors. Among these variables, gestational age, birth weight, body weight at first screening, 1-minute Apgar score, 5-minute Apgar score, and maternal age were treated as continuous variables. The remaining variables were categorical variables. Categorical variables with the same value (e.g., 0 value for all neonates) were removed before data partitioning because these variables could not contribute to model performance. Additionally, if these variables did not occur during the initial screening, bias toward other variables would need not be considered. The remaining variables were used for the

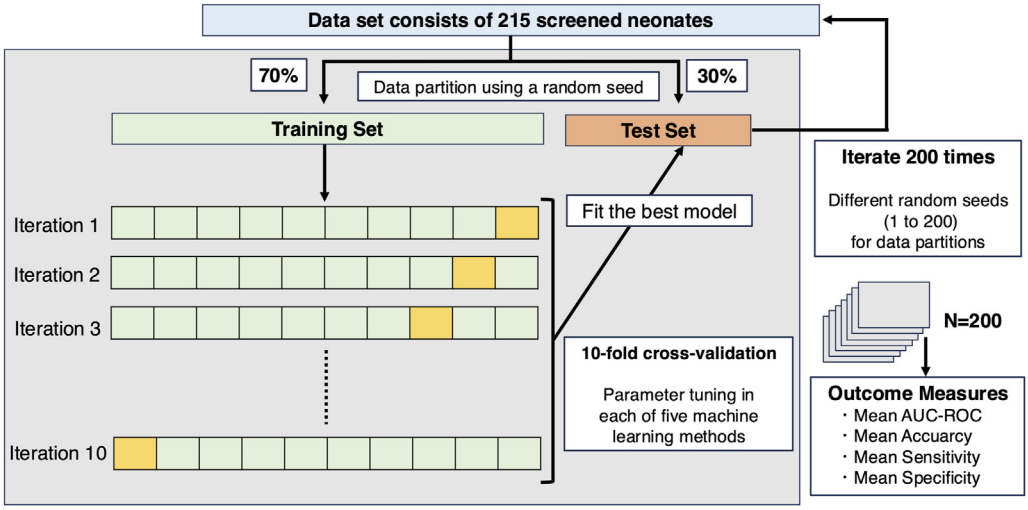


Figure 1. Model development and evaluation algorithm in this study. The 10-fold crossvalidation was used for parameter tuning and the resulting model selection, and the process was repeated 200 times using 200 different random seeds for the data partitioning processes. AUC-ROC = area under the receiver operating characteristic curve.

machine learning processes. A test set was used to assess the validity of the model. The study used the 10-fold cross-validation in the model selection process and the 200-iteration machine learning method to reduce overfitting and bias.

Data Analysis and Prediction Methods

R software version 4.3.1 (R Foundation for Statistical Computing) was used for data analysis alongside model development utilizing the following packages: caret (version 6.0.94),[19] rpart (version 4.1.19),[20] randomForest (version 4.7.1.1),[21] xgboost (version 1.7.5.1),[22] nnet (version 7.3.19),[23] klaR (version 1.7.2),[24] pROC (version 1.18.4),[25] PRROC (version 1.3.1),[26] binom package (version 1.1-1.1),[27] and iml package (version 0.11.3).[28] The caret package was used to organize the 5 machine learning packages: rpart for the decision tree, randomForest for the random forest, xgboost for the gradient-boosted tree, nnet for the neural network, and klaR for the Naive Bayes. The variable importance was calculated using the *varImp* function of the caret package. However, since the Naive Bayes model assumes that each variable is independent, the variable importance could not be calculated for this model. Therefore, the calculation of variable importance was applied to the other 4 models. Similarly, SHapley Additive exPlanations (SHAP) values were computed using the iml package for the 4 models excluding the Naive Bayes model. Confusion matrices were produced using the *confusionMatrix* function of the caret package. The pROC package was used to visualize receiver operating characteristic (ROC) curves and calculate the area under the receiver operating characteristic curve (AUC-ROC). The PRROC package was employed to generate precision-recall (PR) curves and calculate the area under the precision-recall curve (AUC-PR). The mean ROC and PR curves of 200 iterations were created using a nonparametric method.[29] The confidence intervals (CIs) of sensitivity and specificity for a single fit (noniterative) were calculated using the binom package. Unspecified settings were set to the default configurations of the care package.

Figure 1 presents an overview of the algorithms used in this study. The dataset was randomly divided into a training set

Table 1. ROP Diagnosis and Background Characteristics at the First Screening

	Overall (N = 215)	Any ROP (n = 43)	No ROP (n = 172)	*P* Value
Neonatal factors				
Median [IQR] gestational age (wks)	31.4 [28.1–33.4]	25.6 [24.4–27.6]	32.3 [30.4–33.9]	<0.001
Median [IQR] birth weight (g)	1502 [967–1823]	751 [586–903]	1668 [1301–1916]	<0.001
Median [IQR] body weight at the first screening (g)	1566 [1127–1974]	907 [807–1185]	1720 [1356–2026]	<0.001
Intrauterine growth retardation	40 (18.6)	8 (18.6)	32 (18.6)	1
Female sex	104 (48.4)	18 (41.9)	86 (50.0)	0.43
Median [IQR] 1-min Apgar score	7.0 [4.0–8.0]	4.0 [1.0–6.0]	7.0 [5.0–8.0]	<0.001
Median [IQR] 5-min Apgar score	8.0 [7.0–9.0]	6.0 [4.0–8.0]	8.0 [7.8–9.0]	<0.001
Endotracheal intubation	100 (46.5)	40 (93.0)	60 (34.9)	<0.001
Surfactant administration	88 (40.9)	35 (81.4)	53 (30.8)	<0.001
Respiratory distress syndrome	95 (44.2)	38 (88.4)	57 (33.1)	<0.001
Transient tachypnea of the newborn	47 (21.9)	2 (4.7)	45 (26.2)	0.002
Caffeine administration	122 (56.7)	33 (76.7)	89 (51.7)	0.005
Chronic lung disease	0 (0)	0 (0)	0 (0)	NA
Hydrocephalus	3 (1.4)	3 (7.0)	0 (0.0)	0.008
Periventricular leukomalacia	6 (2.8)	6 (14.0)	0 (0.0)	<0.001
Intraventricular hemorrhage	13 (6.0)	10 (23.3)	3 (1.7)	<0.001
Sepsis	10 (4.7)	5 (11.6)	5 (2.9)	0.029
Necrotizing enterocolitis	3 (1.4)	3 (7.0)	0 (0.0)	0.008
PDA	187 (87.0)	42 (97.7)	145 (84.3)	0.038
PDA treated with indomethacin	62 (28.8)	32 (74.4)	30 (17.4)	<0.001
PDA treated with surgery	0 (0)	0 (0)	0 (0)	NA
Erythropoietin administration	105 (48.8)	26 (60.5)	79 (45.9)	0.13
Blood transfusion	28 (13.0)	20 (46.5)	8 (4.7)	<0.001
Human milk feed	214 (99.5)	43 (100.0)	171 (99.4)	1
Maternal factors				
Median [IQR] maternal age (yrs)	32.0 [29.0–36.0]	32.0 [28.0–33.5]	33.0 [29.0–36.0]	0.09
Placental abruption	10 (4.7)	2 (4.7)	8 (4.7)	1
Premature rupture of the membranes	60 (27.9)	9 (20.9)	51 (29.7)	0.34
Placenta previa	16 (7.4)	2 (4.7)	14 (8.1)	0.75
Gestational diabetes mellitus	23 (10.7)	2 (4.7)	21 (12.2)	0.18
Pregestational diabetes mellitus	4 (1.9)	2 (4.7)	2 (1.2)	0.18
Pregnancy-induced hypertension	30 (14.0)	2 (4.7)	28 (16.3)	0.051
Chorioamnionitis	23 (10.7)	11 (25.6)	12 (7.0)	0.001
Steroid administration at delivery	103 (47.9)	28 (65.1)	75 (43.6)	0.019
Other factors				
Cesarean section	168 (78.1)	37 (86.0)	131 (76.2)	0.23
Multiple births	57 (26.5)	7 (16.3)	50 (29.1)	0.13

Apgar = appearance, pulse, grimace, activity, and respiration; IQR = interquartile range; NA = not available; PDA = patent ductus arteriosus; ROP = retinopathy of prematurity.
Data are presented as n (%) or otherwise indicated.

Table 2. Single-Fit Model Performances

Method	AUC-ROC (95% CI)	Accuracy (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
Decision tree	0.94 (0.89–0.99)	84.1 (72.7–92.1)	93.6 (82.5–98.7)	56.2 (29.9–80.2)
Random forest	0.97 (0.93–1.00)	92.1 (82.4–97.4)	96.0 (86.3–99.5)	76.9 (46.2–95.0)
Gradient-boosted tree	0.91 (0.81–1.00)	88.9 (78.4–95.4)	95.8 (85.7–99.5)	66.7 (38.4–88.2)
Neural network	0.88 (0.77–0.99)	87.3 (76.5–94.4)	92.2 (81.1–97.8)	66.7 (34.9–90.1)
Naive Bayes	0.97 (0.93–1.00)	93.7 (84.5–98.2)	98.0 (89.1–99.9)	78.6 (49.2–95.3)

AUC-ROC = area under the receiver operating characteristic curve; CI = confidence interval.

(70%) and a test set (30%) with ROP probability using a specified random seed in R. We used 5 major machine learning methods: decision tree,[30] random forest,[31] gradient-boosted tree,[32] neural network,[33] and Naive Bayes.[34] In the preprocessing step, Min-Max normalization to the continuous variables was applied. Parameter tuning was performed using 10-fold cross-validation on the training set with the following parameters specified in each package: *cp* for decision tree; *mtry* for random forest; *eta*, *max_depth*, *gamma*, *colsample_bytree*, *min_child_weight*, *subsample*, and *nrounds* for the gradient-boosted tree; *size* and *decay* for the neural network; and *usekernel* for the Naive Bayes. The best-performing model was selected based on the accuracy of fit to the test set. To address the uncertainty related to data partitioning using only 1 specified random seed, the data partitioning and subsequent model development process were iteratively repeated 200 times using different random seeds and the *for loop* function in R.

To compare background factors (predictors) between Any ROP and No ROP groups, the chi-square test was applied for categorical variables, whereas the Wilcoxon rank-sum test was utilized for continuous variables.

Outcome Measures

The model's performance was evaluated using the AUC-ROC, accuracy, sensitivity, and specificity as the main outcome measures. The main outcome measures of a single fit (no iterations) and 200-iterate mean were produced.

Results

Summary Statistics

In total, 215 neonates were screened during the study period, including 111 males (51.6%) and 104 females (48.4%). None of the neonates had ROP at the first screening period. The median (interquartile range) gestational age was 31.4 (28.1–33.4) weeks, and the median (interquartile range) birth weight was 1502 (967–1823) g.

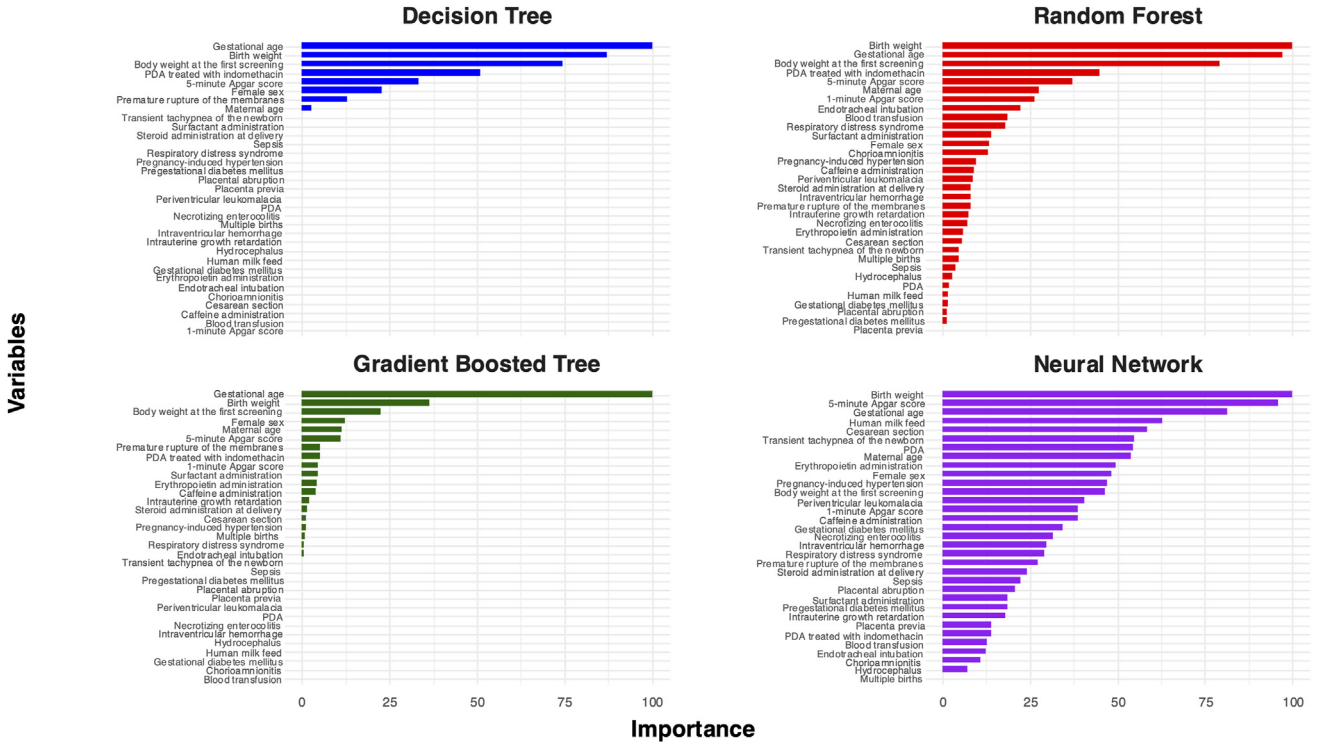


Figure 2. Variable importance of 4 models (single fit without 200 iterations). The single-fit variable importance was calculated using the *varImp* function of the caret package. PDA = patent ductus arteriosus.

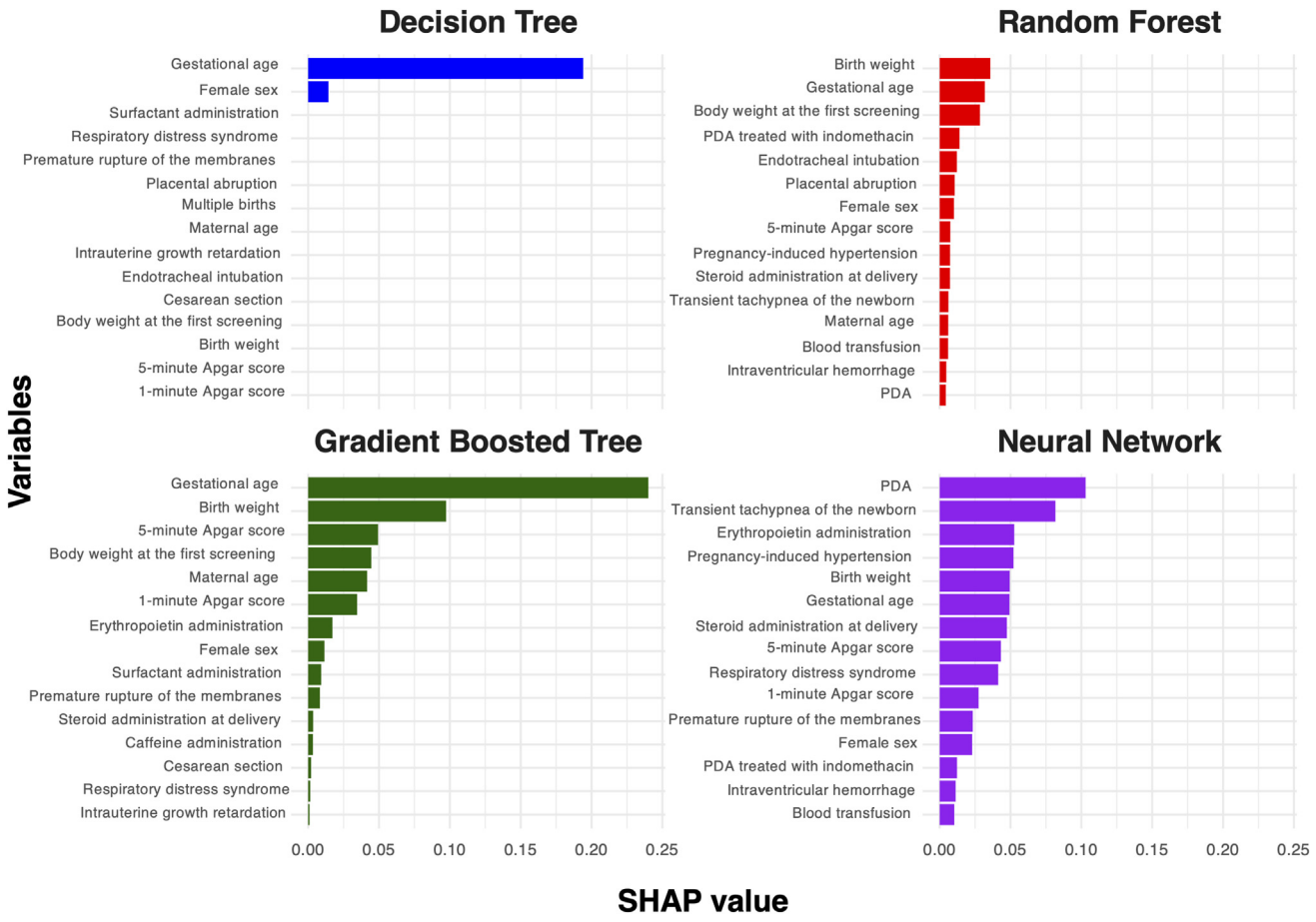


Figure 3. SHAP values of 4 models (single fit without 200 iterations). The single-fit SHAP values were calculated using the iml package. PDA = patent ductus arteriosus; SHAP = SHapley Additive exPlanations.

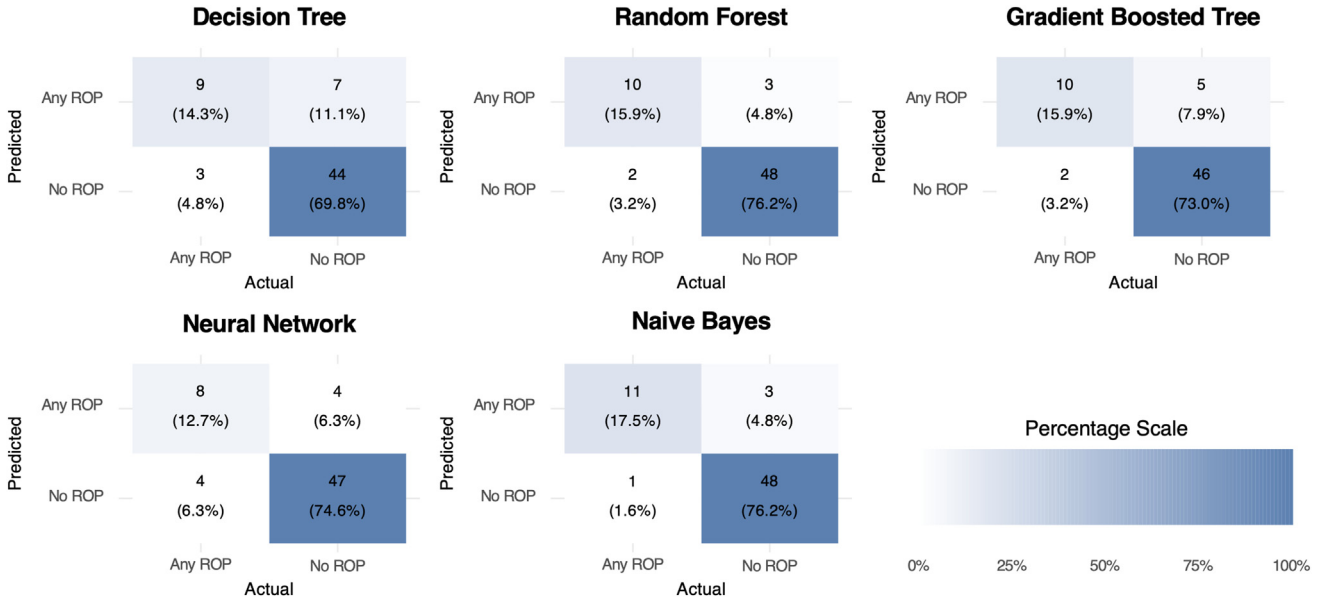


Figure 4. Confusion matrices of all 5 models (single fit without 200 iterations). The single-fit confusion matrices were calculated using the *confusionMatrix* function of the caret package. The color bar in the bottom right indicates the percentage scale. ROP = retinopathy of prematurity.

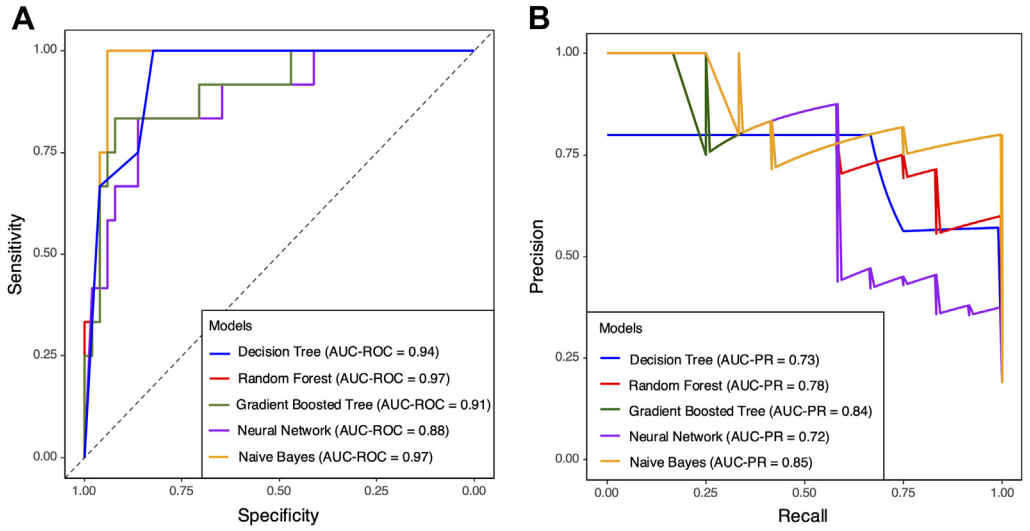


Figure 5. ROC and PR curves of all 5 models (single fit without 200 iterations). The single-fit ROC curves (**A**) and PR curves (**B**) were visualized using the pROC and PRROC packages. AUC-PR = area under the precision-recall curve; AUC-ROC = area under the receiver operating characteristics curve; PR = precision-recall; ROC = receiver operating characteristic.

In this cohort, 43 neonates (20.0%) developed ROP. There are no missing values in this dataset (Table 1).

Preprocessing

As shown in Table 1, 2 neonatal factors (categorical variables), chronic lung disease and patent ductus arteriosus treated with surgery, had zero values. These 2 variables were removed before data partitioning, and the remaining 33 variables were used as predictors.

Model Performance

Table 2 lists the model performance for a single cycle without iterations. The decision tree model produced an AUC-ROC of 0.94, an accuracy of 84.1%, a sensitivity of 93.6%, and a specificity of 56.2%. The random forest model achieved an AUC-ROC of 0.97, an accuracy of 92.1%, a sensitivity of 96.0%, and a specificity of 76.9%. The gradient-boosted tree model showed an AUC-ROC of 0.91, an accuracy of 88.9%, a sensitivity of 95.8%, and a specificity of 66.7%. The neural network model had an AUC-ROC of 0.88, an accuracy of 87.3%, a sensitivity of 92.2%, and a specificity of 66.7%. The Naive Bayes model achieved an AUC-ROC of 0.97, an accuracy of 93.7%, a sensitivity of 98.0%, and a specificity of 78.6%. Figure 2 shows the variable importance, and Figure 3 visualizes the SHAP values of the single fit without iterations. Figure 4 indicates the confusion matrices, and Figure 5 represents the ROC and PR curves.

Table 3 summarizes the model performances over 200 iterations with mean AUC-ROC, accuracy, sensitivity, and specificity. In addition to the main outcome measures, positive predictive value and negative predictive value were also produced: 92.6% (95% CI: 86.2−100) and 69.8% (95% CI: 47.0−100) for the decision tree model; 92.4% (95% CI: 87.5−98.0) and 79.9% (95% CI: 58.3−100) for the random forest model; 92.4% (95% CI: 87.3−97.8) and 71.1% (95% CI: 52.6−91.9) for the gradient-boosted tree model; 91.9% (95% CI: 85.9−98.0) and 73.4% (95% CI: 50.0−100) for the neural network model; and 93.9% (95% CI: 88.7−98.0) and 77.7% (95% CI: 57.1−100) for the Naive Bayes model. Figure 6 represents the variable importance, and Figure 7 visualizes the SHAP values of the 200 iterations. Figure 8 demonstrates the mean 200-iteration confusion matrices,

Table 3. 200-Iteration Model Performances

Method	AUC-ROC (95% CI)	Accuracy (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
Decision tree	0.83 (0.66−0.96)	87.2 (79.4−93.7)	91.9 (80.4−100)	67.4 (33.3−100)
Random forest	0.93 (0.83−0.99)	90.1 (84.1−95.2)	95.7 (88.2−100)	66.0 (41.7−91.7)
Gradient-boosted tree	0.91 (0.81−0.98)	88.1 (82.5−93.7)	93.1 (82.4−98.1)	66.7 (41.7−91.7)
Neural network	0.90 (0.71−0.98)	88.3 (81.0−93.7)	94.0 (86.2−100)	64.1 (33.3−91.7)
Naive Bayes	0.94 (0.86− 0.99)	90.6 (84.1−96.8)	94.6 (86.3−100)	73.6 (50.0−91.7)

AUC-ROC = area under the receiver operating characteristic curve; CI = confidence interval.

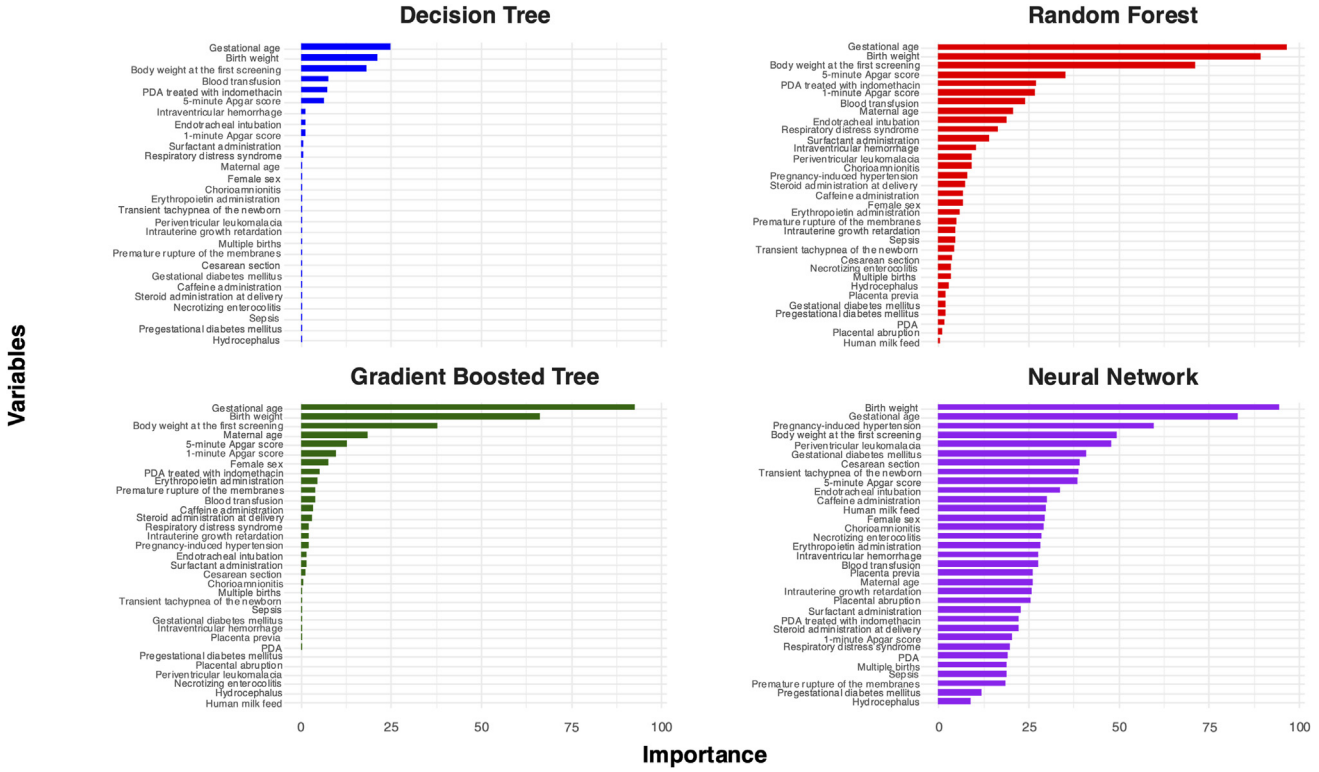


Figure 6. Mean variable importance of 4 models (200 iterations). The 200-iteration mean variable importance was calculated using the *varImp* function of the caret package. PDA = patent ductus arteriosus.

and Figure 9 shows the mean 200-iteration ROC and PR curves using a nonparametric method.

Discussion

The application of artificial intelligence using machine learning models in ophthalmology has recently been reported, particularly for diabetic retinopathy, glaucoma, and ROP. These new techniques have the potential benefit of enhancing diagnostic accuracy and reducing clinical and financial burdens.[35–38] The promising benefits of the techniques in ROP can directly lead to a reduction in childhood blindness worldwide.

Two key areas must be considered when developing solutions for ROP: the occurrence of ROP and the development of treatment-warranted ROP. The former is based on accurate diagnosis, whereas the latter suggests the best timing for intervention. This multicenter observational study, conducted in Yamagata Prefecture, Japan, aimed to develop prediction models for ROP occurrence using nonimaging tabular data. We applied 5 machine learning methods—decision tree, random forest, gradient-boosted tree, neural network, and Naive Bayes—to analyze 215 screened neonates. Based on the 200-iteration results, the 5 machine learning models presented promising results that outperformed prior models despite not utilizing fundus photographs.

Prior research has primarily focused on prediction models that utilize logistic regression, scoring-based models, weight change algorithms, or deep learning models using fundus photographs to predict ROP occurrence. Although some of the first 3 models demonstrated excellent performance in predicting treatment-warranted ROP, they only showed moderate performance in predicting the occurrence of ROP (AUC-ROC: 0.72–0.89).[10–13] The deep learning model achieved high performance in predicting ROP occurrence (AUC-ROC: 0.94); however, it requires fundus photographs captured by pediatric fundus cameras.[14] Therefore, deep learning models using fundus photographs would not be available without pediatric fundus cameras in NICUs. The 4 NICUs in our region did not have pediatric fundus cameras. Moreover, the photographic procedure still physically burdens neonates, although the burden is less than that of ophthalmoscopic fundus examinations. Our machine learning models do not require fundus photographs or the related burden of capturing images.[39] Our study adds to the growing evidence that nonimaging machine learning models can reliably predict ROP occurrence.

The high performance of our models may be attributed to the robust predictor selection and iterative testing processes. Given the availability and lower cost of collecting nonphotographic (tabular) data, our findings offer valuable insights into selecting prediction models. In addition to identifying which model works best, there are

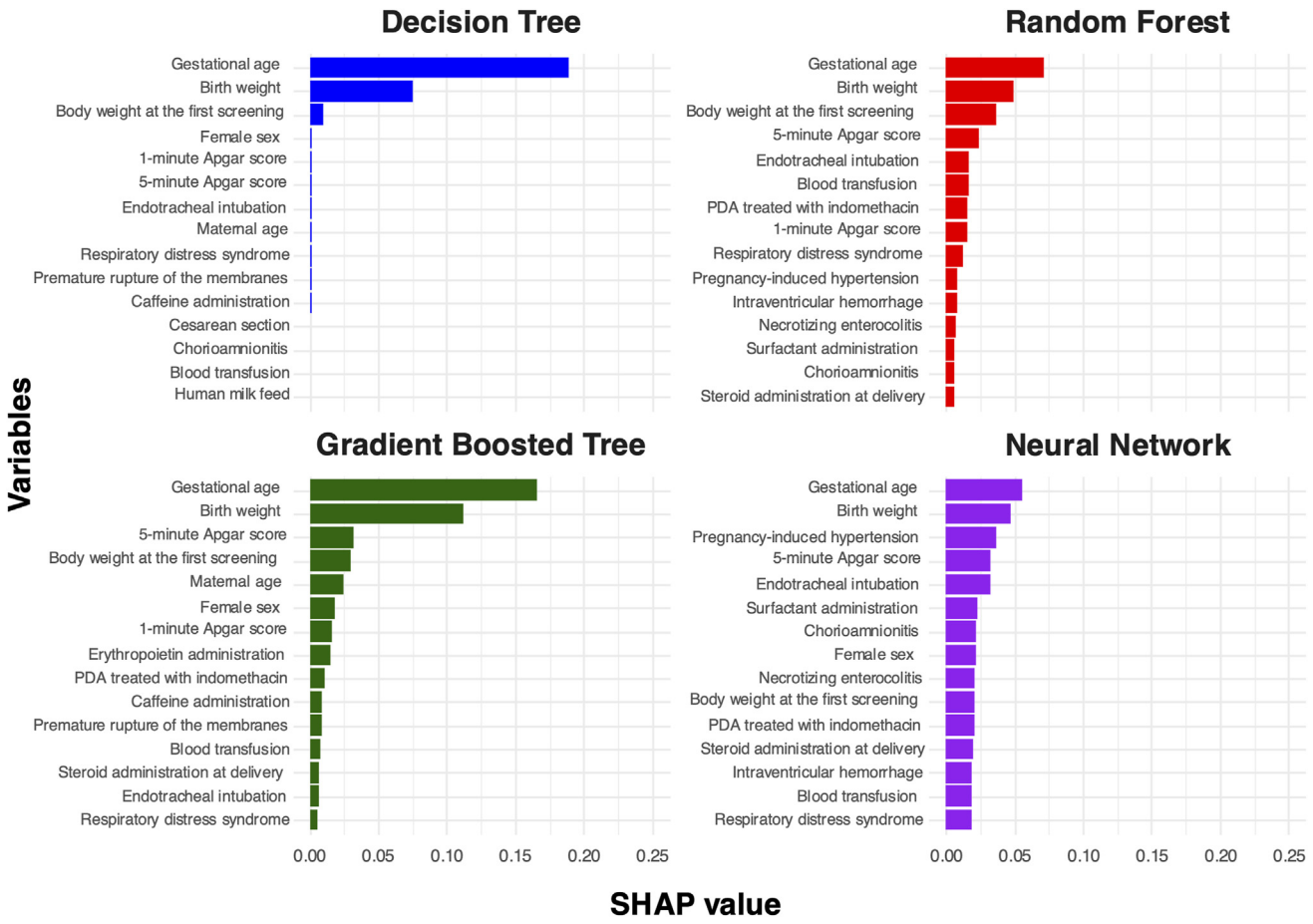


Figure 7. Mean SHAP values of 4 models (200 iterations). The 200-iteration mean SHAP values were computed using the iml package. PDA = patent ductus arteriosus; SHAP = SHapley Additive exPlanations.

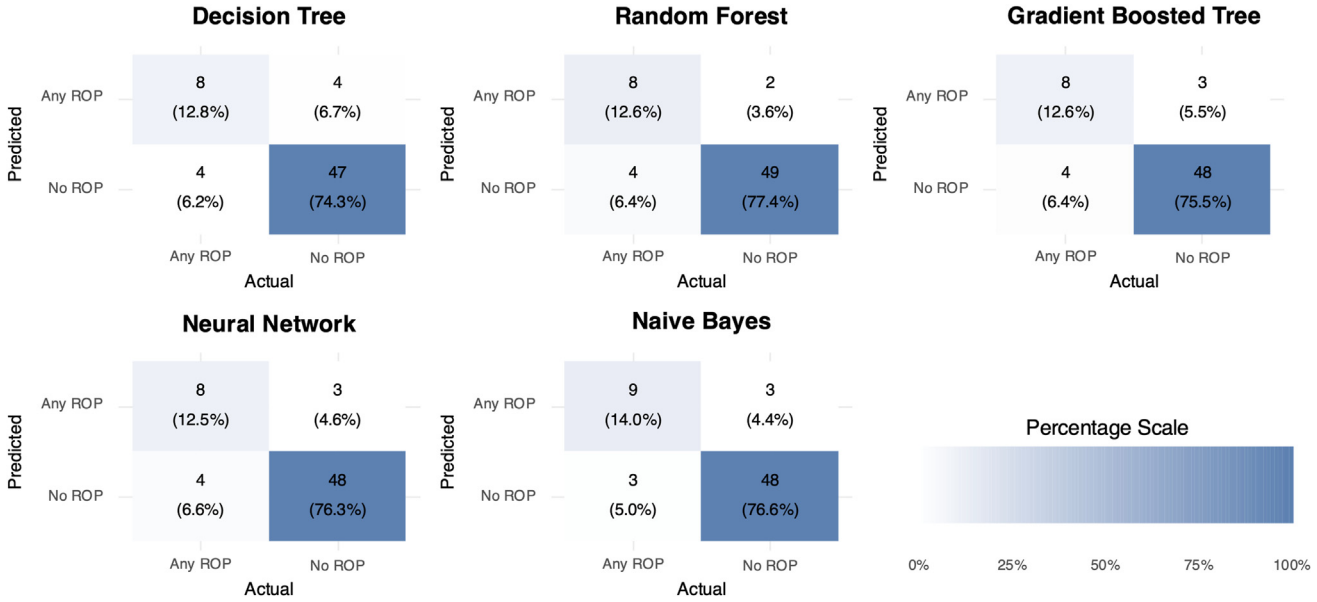


Figure 8. Mean confusion matrices of all 5 models (200 iterations). The 200-iteration mean confusion matrices were calculated using the *confusionMatrix* function of the caret package. The color bar in the bottom right represents the percentage scale. ROP = retinopathy of prematurity.

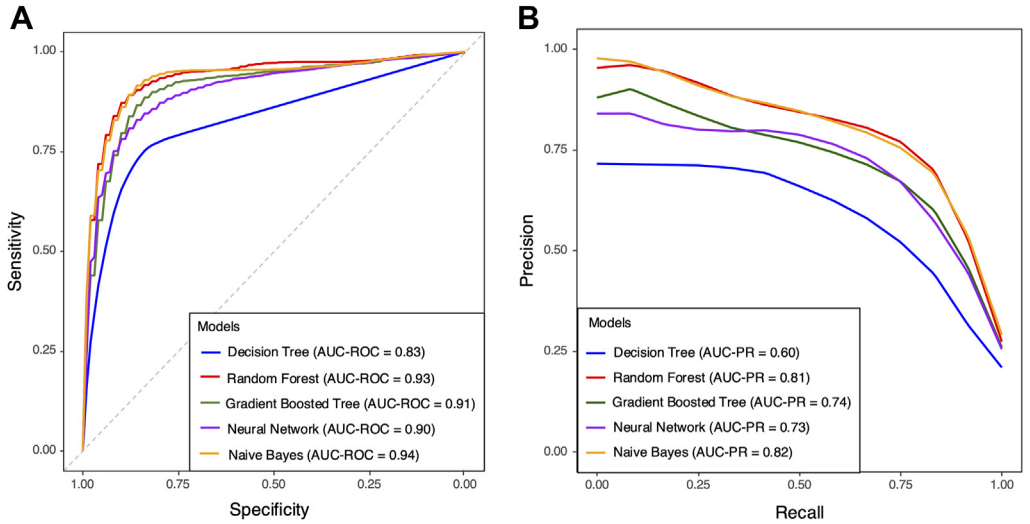


Figure 9. Mean ROC and PR curves of all 5 models (200 iterations). The 200-iteration mean ROC curves (**A**) and PR curves (**B**) were produced using the non-parametric method with the pROC and PRROC packages. AUC-PR = area under the precision-recall curve; AUC-ROC = area under the receiver operating characteristics curve; PR = precision-recall; ROC = receiver operating characteristic.

considerations regarding which model is most practical in different regions or countries and unique cases. The decision may depend on the health care costs and access to medical services in these areas. For example, medical fields without pediatric fundus cameras cannot utilize deep learning models using fundus photographs, and other prediction models may need to be explored. Furthermore, for neonates with severe corneal opacities, prediction models without fundus photographs may be useful because of the difficulty in capturing clear images. These considerations may lead to the development of telemedicine systems.

Our study had several limitations. The sample size of 215 was relatively small for machine learning approaches, which might affect model performance and generalizability. The 10-fold cross-validation within the training set for parameter tuning was used, which may also limit the generalizability of our results. In addition, the potential impact of multicollinearity among certain variables, such as birth weight and gestational age, may affect the model's interpretability and limit its clinical applicability in some settings. Because of concerns about the impact of outliers, determining appropriate cluster numbers, and reduced interpretability of individual variables, we selected not to conduct a preprocessing method such as bottom-up hierarchical clustering, which could potentially improve the multicollinearity in this study. Nevertheless, the random forest model, gradient-boosted model, and Naive Bayes model achieved high AUC-ROC with relatively narrow 95% CIs. Additionally, this study was region-specific and focused only on Yamagata Prefecture, Japan. Geographical limitations may affect the applicability of our findings to other populations. Although the 35 variables used in this study were routinely collected in our neonatal care units, we recognize that their availability may vary in other regions or health care systems. Future research should focus on developing simplified models with fewer variables to maintain high performance while reducing the data collection burden and enhancing generalizability and practical applicability. Furthermore, the Japanese screening criteria (birth weight $\leq$1800 g or gestational age $\leq$34 weeks) have a wider safety margin than that of other countries (e.g., birth weight $\leq$1500 g or gestational age $\leq$30 weeks in the United States[16] and birth weight $<$1501 g or gestational age $<$32 weeks in the United Kingdom).[40] As a result, relatively larger low-birth-weight infants or preterm infants were included; therefore, further validation is needed before it can be applied to neonates screened based on other screening criteria. Finally, our study limits the prediction time for the first fundus examination. Early predictions may contribute to safety management but provide low-performance models owing to fewer predictors. Later predictions may improve the prediction performance; however, they can miss the early development of the ROP. The first fundus examination is a practical time point for predicting the occurrence of ROP.

This study suggests the feasibility of using nonimaging machine learning approaches to predict the occurrence of ROP during the first screening examination. Despite limitations in sample size, geographical area, and screening criteria, these findings have important implications for ophthalmologists and pediatricians interested in easily applicable and robust ROP prediction. These models are promising tools for clinical situations in which clinicians cannot utilize fundus cameras and for future research as prediction models with less human and machine workloads.

Acknowledgments

The authors would like to thank all the included babies and their parents, neonatologists, ophthalmologists, and NICU nurses in Yamagata Prefecture, Japan. We are very grateful to Yosuke Sudo, Satoru Aiba, Kazuhiro Akaba, and Hiroshi Yoshida for pediatric clinical practice in the NICUs and Jun Sakamoto for ophthalmologic examinations at the Tsuroka Municipal Shonai Hospital.

Footnotes and Disclosures

Originally received: May 8, 2024.
Final revision: December 25, 2024.
Accepted: January 13, 2025.
Available online: January 18, 2025. Manuscript no. XOPS-D-24-00143.

[1] Department of Ophthalmology and Visual Sciences, Yamagata University Faculty of Medicine, Yamagata, Japan.

[2] Yamagata City Institute of Public Health, Yamagata, Japan.

[3] Department of Pediatrics, Yamagata University Faculty of Medicine, Yamagata, Japan.

Disclosure(s):

All authors have completed and submitted the ICMJE disclosures form.

The author(s) have no proprietary or commercial interest in any materials discussed in this article.

This study was supported by the Ministry of Education, Culture, Sports, Science and Technology of Japan (Grants-in-Aid for Scientific Research, No. 20K18335). The sponsor or funding organization had no role in the design or conduct of this research.

Support for Open Access publication was provided by the Ministry of Education, Culture, Sports, Science and Technology of Japan.

HUMAN SUBJECTS: Human subjects were included in this study. Written informed consent was obtained from the parents or guardians of patients. This study was approved by the Ethical Review Committee of Yamagata University Faculty of Medicine (No. 2016-244) and conformed to the tenets of the Declaration of Helsinki.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Takeda, Kaneko, Yamashita, Sasaki, Mitsui

Data collection: Takeda, Sasaki

Analysis and interpretation: Takeda, Sugimoto, Yamashita, Mitsui

Obtained funding: Takeda

Overall responsibility: Takeda, Sugimoto, Yamashita, Mitsui

Abbreviations and Acronyms:
AUC-ROC = area under the receiver operating characteristics curve; **CI** = confidence interval; **NICU** = neonatal intensive care unit; **ROP** = retinopathy of prematurity; **PR** = precision-recall.

Keywords:
Retinopathy of prematurity, Machine learning, Nonimaging, Multicenter study.

Correspondence:
Yusuke Takeda, MD, MPH, Department of Ophthalmology and Visual Sciences, Yamagata University Faculty of Medicine, 2-2-2 Iida-Nishi, Yamagata 990-2332, Japan. E-mail: y-takeda@med.id.yamagata-u.ac.jp.

References

1. Blencowe H, Lawn JE, Vazquez T, et al. Preterm-associated visual impairment and estimates of retinopathy of prematurity at regional and global levels for 2010. *Pediatr Res*. 2013;74:35–49.
2. Kong L, Fry M, Al-Samarraie M, et al. An update on progress and the changing epidemiology of causes of childhood blindness worldwide. *J AAPOS*. 2012;16:501–507.
3. Holmström G, Tornqvist K, Al-Hawasi A, et al. Increased frequency of retinopathy of prematurity over the last decade and significant regional differences. *Acta Ophthalmol*. 2018;96:142–148.
4. Ludwig CA, Chen TA, Hernandez-Boussard T, et al. The epidemiology of retinopathy of prematurity in the United States. *Ophthalmic Surg Lasers Imaging Retina*. 2017;48(7):553–562.
5. Löfqvist C, Andersson E, Sigurdsson J, et al. Longitudinal postnatal weight and insulin-like growth factor I measurements in the prediction of retinopathy of prematurity. *Arch Ophthalmol*. 2006;124:1711–1718.
6. Binenbaum G, Bell EF, Donohue P, et al. Development of modified screening criteria for retinopathy of prematurity: primary results from the postnatal growth and retinopathy of prematurity study. *JAMA Ophthalmol*. 2018;136:1034–1040.
7. Pivodic A, Hård AL, Löfqvist C, et al. Individual risk prediction for sight-threatening retinopathy of prematurity using birth characteristics. *JAMA Ophthalmol*. 2020;138:21–29.
8. Pivodic A, E H. Smith L, Hård AL, et al. Validation of DIGIROP models and decision support tool for prediction of treatment for retinopathy of prematurity on a contemporary Swedish cohort. *Br J Ophthalmol*. 2023;107:1132–1138.
9. Wagner SK, Liefers B, Radia M, et al. Development and international validation of custom-engineered and code-free deep-learning models for detection of plus disease in retinopathy of prematurity: a retrospective study. *Lancet Digit Health*. 2023;5:e340–e349.
10. Eckert GU, Fortes Filho JB, Maia M, Procianoy RS. A predictive score for retinopathy of prematurity in very low birth weight preterm infants. *Eye*. 2012;26:400–406.
11. Piermarocchi S, Bini S, Martini F, et al. Predictive algorithms for early detection of retinopathy of prematurity. *Acta Ophthalmol*. 2017;95:158–164.
12. Ricard CA, Dammann CEL, Dammann O. Screening tool for early postnatal prediction of retinopathy of prematurity in preterm newborns (STEP-ROP). *Neonatology*. 2017;112:130–136.
13. Coşkun Y, Dalkan C, Yabaş Ö, et al. A predictive score for retinopathy of prematurity by using clinical risk factors and serum insulin-like growth factor-1 levels. *Int J Ophthalmol*. 2017;10:1722–1727.
14. Wu Q, Hu Y, Mo Z, et al. Development and validation of a deep learning model to predict the occurrence and severity of retinopathy of prematurity. *JAMA Netw Open*. 2022;5:e2217447.
15. Uemura Y, Tsukahara I, Nagata M, et al. "Diagnostic and treatment criteria for retinopathy of prematurity" [translated from Japanese.]. *J Japan Ophthalmol Assoc*. 1975;46:553–559.
16. Fierson WM. American academy of pediatrics section on ophthalmology, American academy of ophthalmology, American association for pediatric ophthalmology and strabismus, American association of certified orthoptists. Screening examination of premature infants for retinopathy of prematurity. *Pediatrics*. 2013;131:189–195.
17. International committee for the classification of retinopathy of prematurity. The international classification of retinopathy of prematurity revisited. *Arch Ophthalmol*. 2005;123:991–999.
18. Early Treatment For Retinopathy Of Prematurity Cooperative Group. Revised indications for the treatment of retinopathy of prematurity: results of the early treatment for retinopathy of

prematurity randomized trial. *Arch Ophthalmol*. 2003;121:1684−1694.
19. caret: Classification and regression training. https://cran.r-project.org/web/packages/caret/index.html. Accessed August 18, 2023.
20. rpart: recursive partitioning and regression trees. https://cran.r-project.org/web/packages/rpart/index.html. Accessed August 18, 2023.
21. randomForest: Breiman and cutler's random forests for classification and regression. https://cran.r-project.org/web/packages/randomForest/index.html. Accessed August 18, 2023.
22. xgboost: Extreme gradient boosting. https://cran.r-project.org/web/packages/xgboost/index.html. Accessed August 18, 2023.
23. nnet: feed-forward neural networks and multinomial log-linear models. https://cran.r-project.org/web/packages/nnet/index.html. Accessed August 18, 2023.
24. klaR: classification and visualization. https://cran.r-project.org/web/packages/klaR/index.html. Accessed August 20, 2024.
25. pROC: display and analyze ROC curves. https://cran.r-project.org/web/packages/pROC/index.html. Accessed August 18, 2023.
26. PRROC. Precision-recall and ROC curves for weighted and unweighted data. https://cran.r-project.org/web/packages/PRROC/index.html. Accessed December 16, 2024.
27. binom: Binomial confidence intervals for several parameterizations. https://cran.r-project.org/web/packages/binom/index.html. Accessed January 15, 2024.
28. iml: interpretable machine learning. https://cran.r-project.org/web/packages/iml/index.html. Accessed December 15, 2024.
29. Chen W, Samuelson FW. The average receiver operating characteristic curve in multireader multicase imaging studies. *Br J Radiol*. 2014;87:20140016.
30. Breiman L, Friedman J, Stone CJ, Olshen RA. *Classification and Regression Trees*. Belmont, CA: Taylor & Francis; 1984.
31. Breiman L. Random forests. *Mach Learn*. 2001;45:5−32.
32. Friedman JH. Greedy function approximation: a gradient boosting machine. *Ann Stat*. 2001;29:1189−1232.
33. Cross SS, Harrison RF, Kennedy RL. Introduction to neural networks. *Lancet*. 1995;346:1075−1079.
34. López Puga J, Krzywinski M, Altman N. Points of significance: Bayes' theorem. *Nat Methods*. 2015;12:277−278.
35. Du XL, Li WB, Hu BJ. Application of artificial intelligence in ophthalmology. *Int J Ophthalmol*. 2018;11:1555−1561.
36. Ting DSW, Pasquale LR, Peng L, et al. Artificial intelligence and deep learning in ophthalmology. *Br J Ophthalmol*. 2019;103:167−175.
37. Benet D, Pellicer-Valero OJ. Artificial intelligence: the unstoppable revolution in ophthalmology. *Surv Ophthalmol*. 2022;67:252−270.
38. Chu Y, Hu S, Li Z, et al. Image analysis-based machine learning for the diagnosis of retinopathy of prematurity: a meta-analysis and systematic review. *Ophthalmol Retina*. 2024;8:678−687.
39. Moral-Pumarega MT, Caserío-Carbonero S, De-La-Cruz-Bértolo J, et al. Pain and stress assessment after retinopathy of prematurity screening examination: indirect ophthalmoscopy versus digital retinal imaging. *BMC Pediatr*. 2012;12:132.
40. Wilkinson AR, Haines L, Head K, Fielder AR. UK retinopathy of prematurity guideline. *Early Hum Dev*. 2008;84:71−74.